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PUBLIC INPUT ON IMPROVING TECHNOLOGY TO EMPOWER MEDICARE BENEFICIARIES

Chags Health Information Technology, LLC (C-HIT), a woman-owned enterprise, offers the following response to the Centers for Medicare & Medicaid Services (CMS) Request for Information (RFI) on the Health Technology Ecosystem. As an organization led by an executive team that comprises a practicing physician as the Chief Executive Officer (CEO), a senior pharmacist as the Chief Administrative Officer (CAO), and Health IT and Federal contracting subject matter experts, along with ISO 9001, ISO/IEC 20000, and ISO/IEC 27001 certifications and CMMI Level 5 process maturity appraisals in development and services, C-HIT provides both health IT and provider clinical services. We deliver innovation that accelerates outcomes by making federal missions more efficient, productive, and secure.

C-HIT was founded in 2013 in support of Affordable Care Act (ACA) systems and Healthcare.gov. As part of the ACA healthcare improvement initiatives, CMS had started working towards implementing systems that promoted "meaningful use." This term referred to the adoption of Electronic Health Records (EHRs) for managing and exchanging patient information, in lieu of paper records and hard copies/folders that patients used to carry with their medical history, hospital admissions, and medications. C-HIT was a major subcontractor in one of the very early programs at CMS, called Electronic Submission of Medical Documentation (esMD), that used HL7, X12, and other IHE to exchange electronically required medical documentation between providers and Recovery Audit Contractors (RACs) as part of audit and recovery towards overpayment of Medicare claims. For another contract, as the prime, C-HIT implemented the interoperable open source CONNECT gateway for a very large Healthcare management solutions company, based on Federal Health Architecture (FHA). This was part of the infrastructure that enabled the organization to connect to what was then the Nationwide Health Information Network (NHIN) to bring in a longitudinal view of patient information that included both clinical and claims details.

Through its investment in Health360 Inc., C-HIT built SOVRINHEALTH 360™, a care coordination platform that supports value-based care (VBC) organizations, such as Management Service Organizations (MSO), Accountable Care Organizations (ACO), Independent Practice Associations (IPA), medical groups, and health plan companies. Since VBC organizations work on a capitated budget model in providing healthcare services to their patient population, they aim to improve patient experience, reduce overall cost of care, improve the population health, and reduce provider burden. The SOVRINHEALTH 360™ Software-as-a-Service (SaaS) platform's features and functions improve the overall quality and performance of healthcare delivery. The platform improves the overall star rating of organizations by improving key Healthcare Effectiveness Data and Information Set (HEDIS) measures, e.g., hospital readmission rates, health assessment, immunizations, cancer screenings, and remote patient monitoring (RPM). Care team members use the platform to provide real-time alerts and reports of emergency room (ER) and inpatient admissions, enhancing their ability to reach out to patients. The platform uses artificial intelligence (AI) for Population Health Management (PHM) that includes risk stratification, care gap identification and closure, entity matching, and unstructured document formatting. The platform has helped VBC organizations by cutting operational costs and increasing revenue through improved

> C-HIT delivers improvements to how patients, providers, and payers access and interact with information. We are currently providing solutions that drive efficient collaboration in the coordination of care. C-HIT implements solutions that use AI to drive Population Health Management and patient and caregiver engagement. The future of healthcare may evolve into ecosystems driven by AI agents who work with one another. C-HIT envisions a future where AI agents command major aspects of healthcare, but with appropriate and critical human guidance and oversight. From today's chatbots to future brain-computer interfaces, C-HIT is committed to supporting CMS's major objective of healthcare delivery that is wellness and outcome driven.

performance. SOVRINHEALTH 360™ cut labor demands by roughly 30% and improved incentive revenue by over 40%.

Leveraging FHIR integration with multiple EHR systems, such as Epic, Veradigm, eClinicalworks, and Cerner, the platform aggregates patient clinical data. The platform also integrates with health information exchanges (HIEs), such as California's Manifest Medex and Maryland's CRISP. The platform receives ADT (Admit, Discharge and Transfer) notifications for patients' hospital visits through these HIE interfaces. The platform also supports importing bulk clinical data through CDA files and bulk FHIR.

For this response, C-HIT has commented on a select group of questions as reflected below.

B. Patients and Caregivers

1. Patient Needs

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

Based on C-HIT's experience, the most effective health management and care navigation apps offer real-time, longitudinal access to clinical, behavioral, and benefit data in a single interface, integrated via TEFCA and powered by AI. Patients and their caregivers use patient portals or mobile apps to gain access to their medical records. This allows them to gain access to lab results, prescribed medications, reports, and appointments. Some of these apps also provide reminders for any proactive wellness activities, such as immunizations and screenings. In addition, patients use apps that connect to RPM devices for real-time vitals monitoring.

One of the struggles that patients and caregivers have is that they have to log in individually into each of their doctor's EHR systems and laboratory services providers to obtain a longitudinal view. In addition, patients and care givers also use wellness apps to track their lifestyle activities.

a. What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products?

Key things that patients and their caregivers would like to do with apps, such as SOVRINHEALTH 360™, include the following.

a) Receive alerts of a patient's vitals for any abnormality. RPM, through integration with medical devices, allows this to become a reality. A digitally active patient or caregiver is able to take proactive measures to deal with abnormal vitals readings, including an immediate follow-up with the primary care physician (PCP), adjusting lifestyle activities, and taking proper medications.
b) Receive alerts from a patient's ER visits. This feature is particularly helpful to the caregiver for proactive engagement in the post discharge recuperation.
c) Receive medication alerts. These alerts are extremely useful for chronic patients as it becomes challenging to adhere to the medication regimen when there are so many medicines to be administered multiple times daily.
d) Enable chatting with a patient's PCP and entire care team. This feature is very useful to both the patients and their caregivers as it provides an easy way to communicate directly with the care team.

b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future.

A personal assistant, such as SOVRINHEALTH 360™, can be extremely helpful to both healthy and chronic patients. Some of the key health-related assistance rendered by personal assistants include:

a) Advise on lifestyle choices. This includes dieting, exercise, and adverse reactions of medicines.
b) Proactively analyze the medical history and forecast health risks for the future. The medical history includes lab reports, encounters, ER visits, and medications.
c) Set up appointments for consultation with primary and specialty doctors and provide reminders.
d) Analyze any condition that the patient may be experiencing. This helps avoid expensive ER visits, while at the same time allows the patient to take action before a certain condition worsens.

PC-2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

Currently, most individuals do not have seamless access to a unified, longitudinal view of all health information in a single portal. Instead, people find themselves piecing together information from multiple systems such as pharmacy apps, EHR-specific portals, insurance platforms, lab portals, and more. This affects patient engagement, care coordination, and proactive management of chronic conditions.

As a patient, it is essential to have longitudinal access to the medical history across the entire care continuum. Since the patient's journey spans primary, specialty, hospitals, labs, and other clinical settings, it would be of greater value to have all the information in a single place. The information can be made available through mobile applications, thereby delivering and making the details available 24x7x365. In addition, it would be beneficial to have all family members' medical histories accessible in a single location or app, subject to HIPAA-based consent requirements. Currently, it is challenging since patients and their caregivers have to use separate credentials to log into their PCP, specialty care, or hospitals EHR systems to retrieve the health information.

a. If so, what are some examples of benefits it has provided?

An effective app would deliver multiple benefits, such as more rapid decision-making in emergencies when medication and diagnosis histories are readily accessible. Other benefits include better care transitions between primary and specialist providers; more informed conversations when the entire record is available to the patient and caregiver; and avoidance of duplicate testing.

The ability to have all information in one place, for families who have minor children and/or elderly parents, will allow the member to not only care for himself/herself but also act as a caregiver to other family members. Some of the key benefits include the following.

a) Ability to receive alerts about key immunization requirements.
b) Ability to present or share immunization information with schools and doctors as required.
c) Ability to receive alerts when the elder's vitals are abnormal and take appropriate action.

d) Time-saving features, e.g., individuals can get information or complete tasks that would otherwise be performed by office staff. Pre-visit paperwork, for example, may be completed in a patient portal, reducing the time spent in the waiting room on the day of an appointment.
e) A patient can communicate with her/his care team more rapidly.
f) A portal allows users to accelerate tasks like requesting refills, receiving records, or making payments.

b. If not, in what contexts or for what workflows would it be most valuable to use one portal or system to access all such health information?

There are multiple scenarios where a single, unified portal or system would deliver value, including, but not limited to the following:

- Managing multiple chronic conditions (e.g., cancer, diabetes, and hypertension) involving multiple specialists.
- Navigating care transitions (e.g., hospital discharge to primary care follow-up).
- Synchronizing care for children or elderly parents, where caregiver proxy access is vital.
- Participating in value-based or preventative care programs, which rely on visibility into gaps in care and patient adherence.
- Sharing decision-making, where complete visibility into prior recommendations, labs, and imaging is necessary.

c. Were there particular data types, such as x-rays or specific test results, that were unavailable? What are the obstacles to accessing your own or your loved ones' complete health information electronically and using it for managing health conditions or finding the best care (for example, limitations in functionality, user friendliness, or access to basic technology infrastructure)?

Per HIPAA, the CMS Interoperability and Patient Access Final Rule (CMS-9115-F), and the CMS Interoperability and Prior Authorization Final Rule (CMS-0057-F), beneficiaries are entitled to access all their medical history with all the data elements. With that said and while we have no issues in accessing data digitally, the access process has the following limitations:

a) Delivery of large dataset details, such as MRI scans, are challenging to be delivered to mobile devices due to bandwidth limitations, so a patient typically has to either receive a hard copy report or log into the portal and download the results.
b) Information presented is very clinical, which makes it difficult for patients and caregivers to take any follow-up action.
c) Lack of transparency to administrative data such as prior authorization.
d) Lack of social determinants of health (SDOH) data for the patient in a manner that allows patients to share with providers and care managers.
e) Currently patients and caregivers have to literally connect with each of the systems individually to have a holistic view.

Industry needs to remove this burden from the patients. For example, patients should have a “health vault” that becomes their central repository for routinely aggregating data from different sources.

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

We recommend that CMS create a CMS Store that could be available through mobile devices, much like the App Store or Play Store. This could be a marketplace where digital health products companies can publish details of their products and the associated costs.

Further, CMS should leverage TEFCA infrastructure, FHIR-based APIs, and CMS claims datasets to collect real-world data on digital health product performance.

2. Data Access and Integration

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

With an almost 100% adoption of EHR systems to electronically capture patient medical data, it is true that all data except for sensitive behavioral data is available to patients and caregivers. The ONC Cures Act on Information Blocking has further facilitated the availability of clinical data. With that said, it is also true that data, such as vitals and medications, are even more readily available since they can be captured at the patient level, without the need for an EHR system.

In C-HIT's experience, the most readily available and valuable health data to patients and caregivers includes the following:

- Vaccination records
- Medication history via EHR portals and pharmacy apps
- Basic laboratory results via portals like Quest or LabCorp
- Appointment summaries and diagnoses from patient portals like MyChart
- CMS claims summaries via tools like Blue Button 2.0

While useful, these data are often scattered, out of sync, and not delivered in a format that supports proactive care navigation or longitudinal health understanding, particularly for high-risk or underserved populations.

a. What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?

Through FHIR-based interoperable exchange, apps can aggregate and provide access to valuable data, such as lab reports, medications, immunizations, procedures, visits, and more. Challenges are tied to accessing administrative data.

a) Patients struggle to understand coverage details for surgical or other procedures. Specialty practices typically take on the responsibility to check with the insurer on the coverage details and then make the patients aware of the coverage. It takes many days to obtain those details. With a digital ecosystem, almost 100% in its adoption, it will be of significant value, it the prior authorization process is more transparent to involve the patient in the loop and make them aware sooner.

b) Plan selection process is still a very tedious process and patients routinely seek the help of plan navigators. This is another area where AI and chatbots can provide patients with an improved experience in their plan selection process.

b. What are specific sources, other than claims and clinical data, that would be of highest value, and why?

Behavioral and social data may be valuable in exposing underlying causes for clinical anomalies. These data are not available in EHR systems or in claims repositories. They come from a variety of sources, such as community, state, and county databases. To have a true picture of a patient's health status, it becomes vital to bring in these data as well.

c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

a) With ONC's Cures Act, and with the increasing number of Qualified Health Information Networks (QHINs), there is a great opportunity to have a 360 degree view of patient information that can be harnessed for clinical research and population health management.
   - CMS's Blue Button 2.0 is a great example of enabling beneficiaries access to claims and clinical data. This fosters an ecosystem with more software developers harnessing interoperability to provide information access.
   - The increasing adoption of TEFCA and QHIN, provides a real and very practical opportunity for the creation of a nationwide "network of networks" that allows for patient data to be available, subject to HIPAA and consent, as meaningful and actionable evidence.

b) As mentioned earlier, the challenge is to find a way to integrate and harness data from other sources such as community, social, and state databases and bring in the necessary patient information. Facilitating a Federal-State partnership can allow interoperability to be practiced at the community hospitals and centers. In addition, those partnerships can also be harnessed to practice interoperability to bring in SDOH data from various sources.

PC-10. How is the Trusted Exchange Framework and Common Agreement TM (TEFCA ™) currently helping to advance patient access to health information in the real world?

The advent of TEFCA as a guiding framework for policies and technical and network requirements has already fostered an ecosystem that enables information exchange in a secure manner between disparate organizations and systems. TEFCA led to the formation of QHINs. QHINs are the backbone in creating a "network of networks" and are the direct result of the framework in creating a nationwide exchange. Currently TEFCA provides support for various Exchange Purposes (EP) (use-cases or categories) The Individual Access Services EP forms the basis for information exchange for patient access. Through this framework and EP, patients now have the ability to have a complete view of their record in one place, assuming that providers are part of the network.

a. Please provide specific examples.

TEFCA, through the Patient Access Services EP, provides patients 24x7 access to their medical history across their entire care continuum. There are now mobile apps, for example, that allow for lab results to

be directly retrieved from laboratory services providers such as LabCorp and Quest; hospital discharge summaries that are made available to the user immediately after a hospital discharge; care gap alerts that provide proactive alerts for any pending screenings and immunizations; and medication lists that include medications from PCPs, specialty care providers, and hospitals.

b. What changes would you suggest?

To bolster TEFCA's impact, C-HIT recommends mandating TEFCA participation by all CMS-contracted payers, certified EHR vendors, and major labs (e.g., LabCorp, Quest), and certified EHR vendors. Certified third-party apps should enable both patient-directed and provider-driven queries. Digital identity workflows could be standardized using NIST 800-63-3 IAL2 CSPs (e.g., Login.gov) to support seamless consent. C-HIT also suggests embedding TEFCA APIs in the CMS Blue Button infrastructure, thereby letting patients access both claims and clinical data in one interface. Finally, CMS could accelerate innovation by funding developer-facing sandboxes and bulk-data testbeds.

d. What standards are you aware of that are currently working well to advance access and existing exchange purposes?

Through experience, C-HIT has identified the following standards that work well:

- FHIR R4 for data structuring and patient-facing APIs
- OAuth 2.0 and OpenID Connect for secure authentication
- C-CDA for backward-compatible document exchange
- IHE XCA/XDS.b for cross-community queries in HIE settings
- SMART on FHIR for app authorization and launch within EHRs

e. What standards are you aware of that are not currently in wide use, but could improve data access and integration?

The following could be used to improve access and integration:

- FHIR subscriptions for real-time updates and event-driven workflows
- Bulk FHIR (Flat FHIR) for population-level insights and multi-patient navigation
- FHIR Questionnaire and QuestionnaireResponse for dynamic intake and assessments
- CDS Hooks to surface context-aware recommendations during care
- Unified Data Access Profiles (UDAPs) for secure, scalable API authorization

f. Are there redundant standards, protocols, or channels that should be consolidated?

The following are redundant channels that could be migrated to TEFCA-based exchange using modern APIs and common security standards:

- DirectTrust/Direct Secure Messaging, which is often siloed and poorly integrated with patient apps
- Point-to-point HL7v2 interfaces, which are costly to maintain and lack patient-facing transparency
- Proprietary EHR APIs, which vary widely and undermine developer scalability

g. Are there adequate alternatives outside of TEFCA for achieving widespread patient access to their health information?

Based on C-HIT's experience, there are no existing viable alternatives that match TEFCA's scalability, national governance, and trust framework to unlock the next generation of patient-centered innovation. TEFCA is best positioned to deliver a nationwide, consent-driven, multi-party data exchange that includes patients, payers, providers, and public health entities.

PC-11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?

HIEs are a direct result of "meaningful use" which advocated the use of EHR systems to capture, store, and retrieve patient clinical data such as medications, lab reports, and encounters. The adoption of EHR systems quickly led to "information silos." The information silos posed a significant challenge in providing a 360 degree view of a patient's medical history across the entire care continuum. HIEs act as the hub that connects either regionally or locally to all of the EHR systems and across various clinical settings (e.g., primary, specialty, hospitals, rehab). The HIEs bridge the gap in bringing in patient data across the entire care continuum. HIEs bring in the following types of data: a) ADTs messages that in real time reflect the patients' emergency and inpatient hospital episodes, as well as specialty visits, and b) CDAs (Clinical Document Architecture) that carry Continuity of Care Document (CCD) information for additional clinical data collected from specialty care and other clinical settings. For example, as a company already certified and tested with the Southern California HIE, Manifest Medex, SOVRINHEALTH 360™ receives hundreds of ADTs and CDA documents from the HIE. This has led to the significant reduction of retrospective patient outreach by alerting the care team in real time of the hospital episodes.

a. How valuable, available, and accurate do you find the data they share to be?

HIEs are generally well connected and usually cover more than 95% of the facilities in each region. They are typically very accurate, since the clinical or administrative data they collect and transmit comes from the source. The information is extremely valuable in providing, in real time, patient ER visits and inpatient admissions to the providers, their staff, and case managers for proactive patient outreach. This allows the VBC organizations to perform and score better against HEDIS and other performance measures.

c. Are there particular examples of high-performing HIE models that you believe should be propagated across markets?

An outstanding example is the Chesapeake Regional Information System for Our Patients (CRISP), the State-Designated HIE and Health Data Utility (HDU) for Maryland and District of Columbia (DC). CRISP facilitates the electronic transfer of clinical information between disparate health information systems. CRISP connects health systems, providers, and various healthcare entities to ensure comprehensive and up-to-date data is available for informed clinical decision making. The CRISP platform aggregates data from multiple sources, including primary care practices, hospitals, and specialty care providers, creating a unified health record that streamlines care coordination and minimizes service duplication. CRISP Shared Services also supports other states like Connecticut, Virginia, West Virginia, and Alaska.

d. What is the ongoing role of HIEs amidst other entities facilitating data exchange and broader frameworks for data exchange (for example, vendor health information networks, TEFCA, private exchange networks, etc.)?

The current HIEs have matured well over the last decade and bring in the experience having resolved many of the bandwidth and data issues. In addition, each of these HIEs has numerous clients who have subscribed to their services. HIEs are essential for whole-person care and population health and can be an integral part of QHIN to promote the next level of interoperability of supporting a "Network of Networks."

3. Information Blocking and Digital Identity

PC-14. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/ AAL2 credentialing service providers (CSP)):

a. What are the challenges today in getting patients/caregivers to sign up and use digital identity credentials?

In C-HIT's experience, challenges to getting patients and caregivers to use digital identity credentials include the following:

- Trust: Individuals may distrust government or commercial identity providers due to concerns about privacy or past data breaches.
- Lack of awareness: Many patients and caregivers are unaware that digital identity credentials exist.
- Difficult onboarding: Verifying identity through CSPs can involve uploading documents, using mobile apps and other steps, that may be difficult for disabled, elderly, or non-English-speaking users.
- Inadequate digital literacy: A significant portion of the Medicare population lacks experience with smartphone apps, two-factor authentication, account recovery methods, and other digital tools.

b. What could be the benefits to patients/caregivers if digital identity credentials were more widely used?

More widespread use of digital credentials could deliver a variety of benefits, including the following:

- Seamless access to personal health records across providers, payers, and apps with one secure login
- Secure consent management for third-party data sharing
- Proxy access for caregivers, with validated relationships and audit trails
- Reduced need for multiple usernames/passwords across different patient portals
- TEFCA exchange is supported, i.e., real-time patient-authorized access to longitudinal records is enabled

c. What are the potential downsides?

We do not see any significant downsides to CSP usage by patients from a security and usage perspective. CSPs can be either for-profit or non-profit organizations. One obvious risk is that a CSP may go out of business. Other potential downsides include the following:

- Without inclusive onboarding options, some patients may be excluded
- Adoption could be delayed by concerns about surveillance, data tracking, or identity misuse

- Credentialing issues, e.g., forgotten passwords or inaccessible devices can prevent patients from accessing their medical information, especially in emergency situations

d. How would encouraging the use of CSPs improve access to health information?

Encouraging the use of CSPs may deliver benefits, such as the following:

- Enabling single-sign-on (SSO) to all TEFCA-connected apps and portals using strong, standardized authentication
- Abridging auditability and accountability for patient-authorized data exchanges
- Reducing friction on the provider side by streamlining real-time consent workflows
- Simplifying caregiver delegation, proxy management, and guardianship workflows across health systems

e. What role should CMS/payers, providers, and app developers have in driving adoption?

The best way to improve adoption of CSPs is to drive the registration and activation process in the app through integration with CSPs. In other words, the apps will enable SSO with CSPs to reduce the burden for the provider community. To improve adoption, CMS could mandate CSP acceptance for certified apps and Medicare-facing systems. Further, CMS could provide grants for Federally Qualified Health Centers (FQHCs) and rural clinics to support in-person identity verification. Finally, CMS could establish digital identity as a core part of meaningful use and TEFCA onboarding requirements.

Payers could improve adoption by auto-enrolling new members in CSP workflows during benefits onboarding; and embedding CSP credentials into member portals and mobile apps.

Providers could improve adoption by offering in-clinic enrollment support; enabling CSP login for patient portals; and educating patients about the benefits of digital identity at the point of care.

Finally, software developers should integrate federated CSP login into onboarding, and build credential recovery and multilingual support into user interfaces.

f. How can CMS encourage patients to get digital identity credentials?

CMS can encourage patients to get digital identity credentials by integrating mymedicare.gov with CSPs, such as ID.ME or Login.gov – similar to IRS and SSA, which have adopted them for their users. This will enable a large-scale adoption of CSP credentials amongst the more than 68 million Medicare beneficiaries who are currently enrolled in Medicare.

CMS can encourage bundled enrollment, i.e., provide digital identity setup as part of Medicare registration online. CMS can embark on multilingual education campaigns, running ads, mailers, and videos, explaining how digital identity enables easier access and better care. In underserved areas, CMS could offer financial or technical assistance.

CMS also could link digital identity usage to faster prior authorization, simplified eligibility checks, or bonus payments under value-based models.

E. Technology Vendors, Data Providers, and Networks

1. Ecosystem

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

CMS had been a pioneer in advancing healthcare delivery modernization through payment reform models. To that effect, CMS has set a key objective to transform the entire Medicare and Medicaid population under a VBC model by attributing the population to an accountable care model. VBC associates significant incentives to providers for reducing cost of care and improving patient experience. VBC is built on the patient engagement factor to achieve the objectives. Beneficiary engagement can be significantly improved if beneficiaries adopt digital health products.

To improve developer interest in the short-term, CMS can stimulate developer interests tactically by

a) Having a metric to measure the effectiveness of plan organizations who bid for Medicare plans in engaging the beneficiaries through adopting digital health products as part of the patient engagement requirements; and
b) Conducting periodic codeathons to allow developers to demonstrate value-add products catering to the following: CMS Interoperability and Prior Authorization Final Rule, Blue Button initiative, and ONC Cures, to name a few.

CMS's long-term strategy should focus on public-private partnerships that goes beyond Medicare and Medicaid. Since performance-based contracts are an increasingly viable business model for MSOs and IPAs who focus on commercial contracts with health plans, CMS stands to benefit by engaging and participating as a lead payer in the nation's digital health product adoption.

TD-2. Regarding CMS Data, to stimulate developer interest—

a. What additional data would be most valuable if made available through CMS APIs?

For organizations like C-HIT, our developers would benefit most from access to real-time lab results, imaging reports, ADT notifications, prior authorization status, and coverage eligibility data through standardized CMS APIs. These data types are essential for powering clinical decision support, care coordination, and patient engagement applications. Other valuable information includes pharmacy data, clinical notes, biometric data, social determinants of health data, and care gap information.

b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API?

To complement Blue Button 2.0, the most valuable data sources include, but are not limited to: clinical data via TEFCA (e.g., EHR-sourced data through QHINs); SDOH data from community organizations; home health records; hospice records; imaging and pathology data; and Medicaid encounter data. These sources provide a more comprehensive, longitudinal view of patient health, enabling the precision targeting of interventions.

c. What obstacles prevent accessing these data sources today?

Obstacles include, but are not limited to, inconsistent data standards, siloed systems, limited real-time access, lack of payer and provider TEFCA participation, siloed data systems with no FHIR API support, proprietary data contracts or non-standard API implementations, inconsistent consent and authorization frameworks, and limited infrastructure for real-time query and event-driven data exchange.

d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

CMS should expand API offerings and align them with TEFCA exchange. This could dramatically increase developer engagement and unlock innovation that benefits patients, providers, and payers.

2. Digital Identity

TD-3. Regarding digital identity implementation:

Digital identity has become ubiquitous in the modern times. It has become part of the mainstream for most aspects of people's daily activities, including finance and healthcare. After the meaningful use introductions as part of ACA, the healthcare industry has adopted EHR at the national level, with data clearly indicating that the adoption of EHR is close to 100%. Digital identities are the thread in the fabric of clinical and administrative data for members. Various types of users access patients' data and that includes patients, their caregivers, doctors and staff, plan companies, and federal agencies such as CMS and NIH. In addition, the Federal Government has embraced digital identities for all its users, including citizens, contractors, vendors, and employees. While adoption of digital identities has increased substantially, the lack of standardization of digital identities has led, and continues to lead, to a lot of waste. Redundant systems and implementations exist, and users have to enroll multiple times and this has led to identity proliferation.

a. What are the challenges and benefits?

Digital identity implementation challenges include privacy concerns, implementation costs, and digital literacy barriers. The benefits of digital identity implementation include enhanced security, reduced fraud, and streamlined access to care.

Digital identities make it convenient, while being secure, to transact with healthcare systems. It makes it easy for Medicare beneficiaries to use services such as Mymedicare.gov, to look for plans, understand benefits and costs of premium and out of pocket expenses, and have secure conversations with doctors and other medical staff. Challenges to maintaining digital identities are in the form of protection to them to ensure compliance with security and privacy requirements; ease of use, while not compromising security; and proliferation due to multiple systems having the requirement for patients to register and receive credentials

b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?

Requiring digital identity credentials, such as CLEAR, would create a more secure and standardized identity ecosystem, ensuring patient information is only accessed by authorized users. While use of digital identity credentials is a natural progression and a mandatory element in the digital health ecosystem to build trust between systems, it also opens the door for more cybersecurity attacks. Credentials must be properly secured to prevent them from being stolen or changed. Digital credentials that are not secured properly allow for either denial of services or improper and fraudulent use of systems and services. It is also important for enterprises to build in security measures and firewalls to protect against such cybersecurity attacks. In addition, users and, in particular, the citizen population have to become aware of security measures and social engineering activities that can result in malicious or inappropriate access to personally identifiable information (PII) and protected health information (PHI) by unauthorized users.

c. What impact would mandatory use of the OpenID Connect identity protocol have?

Use of OpenID should be part of the plan to leverage identity and credential service providers to allow users access to enterprise applications using secure SSO and share information. This opens the path to improved security and convenience while reducing costs of managing identities within enterprises. This also opens the possibility of being very accommodative for future regulation and rule changes in the use of digital identity ecosystem.

3. Technical Standards and Certification

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

CMS can better encourage use of open, standards-based, publicly available APIs by tying reimbursement incentives and certification eligibility to FHIR-based, open APIs. CMS can shift the ecosystem away from proprietary systems. The path to adoption of any standard is in the guidance, rulemaking, and enforcement, enabling businesses working with CMS to adopt public APIs, such as Open ID and FHIR.

Additionally, to incentivize compliance, CMS can link program participation, payment eligibility, and quality bonuses to the demonstrated use of standard APIs. Another strategy would be for CMS to provide innovation grants or technical assistance for developers building open APIs, particularly those addressing underserved populations and interoperability challenges.

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?

Directory services are fundamentally registries that allow for establishing an authoritative source for consumers. A nationwide provider directory of FHIR endpoints allows for an easy way to connect and gather patient data from any clinical setting. This is an extremely useful service that can be leveraged by patients (through the use of mobile apps) for easy lookup in search of specialty doctors in good standing, good hospital, urgent care, etc. A payer directly benefits through improved contracting (for

health plans) with provider organizations and provides them with the ability to pull in data directly from provider systems as part of their claims processing. Providers benefit from direct provider-to-provider (P2P) communication requirements that need exchange of clinical and administrative information. A nationwide provider directory cannot be built by one single organization, but must be built through a collective effort of multiple organizations working in unison to achieve the objective of creating the authoritative source of provider information. CMS, as the leading payer, can publish that authoritative source. A decentralized model where the repository is collectively managed by a community of payers (i.e., an alliance), however, would scale better and be more cost effective through a cost sharing agreement among the participating organizations and partners. Much like CMS's Marketplace, the costs of maintaining and enhancing the directory could come from fees from the members of the alliance, along with a charge to organizations that want to be part of the network. The directory should be available for free for the beneficiaries. This directory will serve as the source of truth for provider and supplier information to all the consuming participants.

TD-6. What unique interoperability functions does TEFCA perform?

TEFCA establishes a common governance framework to allow for healthcare networks to interoperate with each other, effectively creating a "network of networks." The QHINs that are already onboarded by the Recognized Coordinating Entity (RCE) allows for this to become a reality. TEFCA provides technical, infrastructure, and governance for the QHINs. At its core, TEFCA uses FHIR as the interoperable protocol with US Core Data for Interoperability (USCDI) as the data model.

TD-8. What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

Based on C-HIT's experience, the most effective certification criteria and standards under the ONC Health IT Certification Program are the following:

- API interoperability
- Mandatory FHIR-based APIs
- Patient access capabilities
- Testing across real-world use cases
- Ability to integrate with national HIEs
- Clinical decision support integration

TD-9. Regarding certification of health IT:

a. What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

The benefits of prioritizing API-enabled capabilities over software functionality for certification are as follows:

- Innovation is enabled
- Interoperability is improved
- Vendor lock-in is reduced

With SOVRINHEALTH 360™, C-HIT integrates with health plan systems, illustrating how API-first approaches enable flexible, scalable solutions that improve care coordination.

4. Data Exchange

TD-14. Regarding networks' use of FHIR APIs:

a. How many endpoints is your network connected to for patient data sharing? What types, categories, geographies of endpoints do you cover? Are they searchable by National Provider Identifier (NPI) or organizational ID?

We support two types of connection endpoints for patient data sharing – partnered and partnerless. A partnered endpoint is set up through an exclusive partnership arrangement as a business associate. The endpoints are searchable through an Organization ID. In addition, we also support connecting to hundreds of clinics and hospitals that expose their FHIR connection endpoints and are integrated through patient access FHIR APIs.

The partnerless end points are searchable through organization names/IDs. Currently, in our network, we have approximately 20 private endpoints and are connected to hundreds of FHIR endpoints through the partnerless model. We will be expanding the service to include more Managed Service Organizations (MSOs), Managed Care Organizations (MCOs), ACOs, IPAs, and provider health systems to connect to their EHR systems as part of data aggregation.

b. How are these connections established (for example, FHIR (g)(10) endpoints, TEFCA/Integrating the Health Enterprise (IHE) XCA, or proprietary APIs)?

The connections have been established through partnerships with the EHR vendors, having our interfaces tested and certified by the EHR vendors. We have used both proprietary APIs (such as Unity API with Veradigm, formerly Allscripts) and standards REST-based interfaces (FHIR). . For example, in addition to the standard interfaces, we also integrated with Manifest Medex, a HIE based in Southern CA, using SFTP interface to receive ADT notifications for hospital episodes, along with CDA for CCDs. As another example, we have supported IHE XCA interfaces with the interoperability initiative within CMS.

c. Do you interconnect with other networks? Under what frameworks (for example, TEFCA, private agreements)?

All our networks are private networks based on both standards-based and proprietary interfaces.

TD-15. Regarding bulk FHIR APIs:

Bulk FHIR is particularly useful when you are exchanging large volumes of data as part of the medical history sharing for care coordination and sharing information about a cohort of patient population for PHM.

a. How would increased use of bulk FHIR improve use cases and data flow?

Bulk FHIR brings in significant advantages (as opposed to retrieving data from one patient at a time) for VBC organizations to improve their overall performance in managing high-risk population. The encounter, lab reports, procedures, diagnosis, notes and various radiology reports are fundamental for Risk stratification and recommendations for better clinical decision making. This will improve the overall

care coordination workflow and enables the care team to proactively outreach to the patients for follow-ups, medication management, and behavioral and wellness management.

b. What are the potential disadvantages of their use?

Some of the challenges that we see with Bulk FHIR are more implementation centered. Performance and security are two primary concerns that we have since large volume data is being exchanged as part of Bulk FHIR. Bulk FHIR is typically used as a batch process in order to prevent any performance impact on daily operations. Since Bulk FHIR contains the data for multiple patients, the need to conduct accurate demographic data reconciliation with existing data quickly is critical (e.g., name, date of birth, plan number ID). Failure to have an accurate name resolution may lead to a "patient data breach" scenario.

TD-16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?

Point-to-point models will have a significant scalability, administrative, and cost issue. Additionally, there is no standard way to regulate the point-to-point model for auditability and accountability, and it is not amenable to proper oversight on data sharing. A shared network infrastructure provides economies of scale and brings in the right level of regulation and oversight.

a. Do current rules encourage scalable network participation?

The Cures Act, CMS Interoperability and Patient Access Final Rule, and CMS Interoperability and Prior Authorization Final Rule provide the governance and the regulatory oversight over interoperability and information blocking. In addition, the TEFCA framework from the US Department of Health and Human Services' Assistant Secretary for Technology Policy (ASTP) lays all the groundwork for QHINs that makes it easier to be part of the network since all the infrastructure, network, and governance are offered in a SaaS model for provider health systems and payers to use to gain access to longitudinal data.

TD-17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

Through their rule-making process, such as the CMS Interoperability and Patient Access Final Rule, CMS Interoperability and Prior Authorization Final Rule, and Cures Act, CMS can establish guideline factors for a broader adoption of the networks by providers, payers, and other third-party organizations, such as MSO and ACOs. By tying the payment schedule for Medicare- and Medicaid-related services with interoperability goals, CMS will be able to encourage and motivate the increased usage of the data sharing networks. In turn, these outcomes will enable CMS to achieve its objective of shifting from Fee for Service (FFS) to the VBC model.

F. Value-Based Care Organizations

1. Digital Health Adoption

VB-3. What are essential health IT capabilities for value-based care arrangements?

The main goal for any VBC is to support better outcomes by improving the overall care coordination for patients, particularly for those who have chronic conditions in order to achieve the Quadruple Aim of

improving patient experience, reducing costs, reducing provider burden, and improving the population health. Since many VBCs participate in financial-risk models to deliver care, they will have to ensure that their operational and utilization costs are kept under control. To achieve this, VBCs need a very efficient care coordination framework. VBCs can provide a better care coordination model if they build health IT capabilities around hospital discharge follow-ups, hospital re-admission rate reductions, medication management (adherence and compliance), timely patient outreach, and care gap management. To that extent, VBCs are served better if they have capabilities to connect with HIEs for ADT and CDA notifications, provide RPM for real-time vitals management for chronic patients with two or more chronic conditions, AI for care gap management, and PHM.

a. Examples (not comprehensive) may include: care planning, patient event notification, data extraction/normalization, quality performance measurement, access to claims data, attribution and patient ID matching, remote device interoperability, or other patient empowerment tools.

Based on our experience with VBC organizations, C-HIT has identified a range of important health IT capabilities. VBC care benefits from coordination across team members and care settings. Leveraging AI to identify, prioritize, manage, and track care tasks is vital. For example, SOVRINHEALTH 360™ depends on AI to automate routine processes, while relying on healthcare practitioners for oversight and decision-making. SOVRINHEALTH 360™ identifies when patients are missing recommended services or follow-ups, with AI-powered analytics to rank gaps based on clinical impact and automated task creation to address these gaps. With AI, we can deliver comprehensive care planning with shared, dynamic care plans accessible to the patient and all care team members. AI can suggest interventions based on evidence and best practices, while clinicians remain in control of final decisions. Other AI-powered capabilities may include real-time alerts for admissions, discharges, and transfers, while prioritizing follow-up activities based on patient risk and care team availability. Another key capability would be integrating secure messaging with workflow tools, enhanced by AI, to route messages, suggest actions, and facilitate information sharing. We can use SOVRINHEALTH 360™ to retrieve data from diverse sources (e.g., HIEs, EHRs, claims) normalize it, identify patterns, and deliver a unified patient view.

b. What other health IT capabilities have proven valuable to succeeding in value-based care arrangements?

C-HIT has found that organizations often focus heavily on clinical data analytics, while underinvesting in the operational workflow systems needed to actually execute on the insights. To succeed, VBC requires both analytical capabilities and robust workflow management tools that increasingly leverage AI to reduce administrative burden while keeping practitioners in the loop for critical decisions.

3. Technical Standards

VB-11. What specific interoperability challenges have you encountered in implementing value-based care programs?

Through our work with SOVRINHEALTH 360™, C-HIT has observed and addressed interoperability challenges. In creating SOVRINHEALTH 360™, C-HIT recognized the importance of enabling different systems to participate in and trigger actions within a shared, cross-organizational process. A referral initiated in one system, for example, should seamlessly create a task and relevant data packet in the receiving system. With SOVRINHEALTH 360™, C-HIT bridges many workflow gaps, leveraging AI to route, rank, and manage cross-system workflows. In C-HIT's experience, current interoperability limitations

restrict the ability of AI systems to access the comprehensive data needed for optimal functioning. In today's healthcare environment, AI solutions confined to single-system data have limited utility. Other challenges include, but are not limited to, lack of standardized data for care coordination; inconsistent information source quality and timeliness; inability to gather unified, real-time views of patients; and, problems tied to accessing and integrating data from external partners.

VB-15. How could a nationwide provider directory of FHIR endpoints help improve access to patient data and understanding of claims data sources? What key data elements would be necessary in a nationwide FHIR endpoints directory to maximize its effectiveness?

Almost every EHR product has built in the FHIR capability in compliance with the CMS Interoperability and Patient Access Final Rule. A nationwide directory would enable mobile application developers to build capabilities to enable patients to have access to their data without any restrictions on the geographical location of the clinic or hospital they visit. Currently many EHR vendors provide such a directory, but one with information limited to only those clinical setting that have their product deployed. SOVRINHEALTH 360™, through its partnership with EHR vendors, such as Epic, Cerner, Veradigm, AthenaHealth, eCW, and NextGen, has built its own provider directory of FHIR endpoints. This enables our member population to have seamless access to their data across multiple settings and across the nation. In addition, these endpoints, as is the case with Blue Button 2.0 (another FHIR endpoint), allow patients access to benefits and claims related data, such as explanation of benefits (EOB), services and payments made, and out of pocket and co-payment charges.